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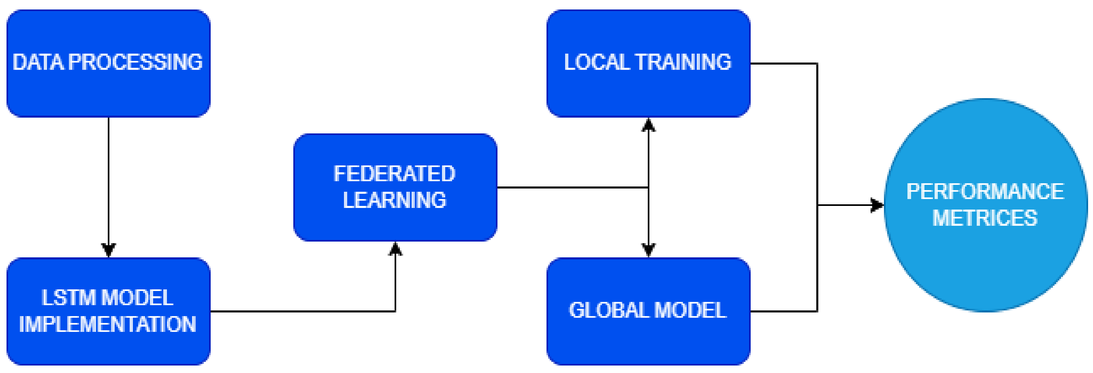
Development of a Federated Multimodal Digital Twin Concept for Industrial IoT

Mohammad Jasarat Anjum

Motivation

- Industry 4.0 generates massive IoT sensor data
- Practical deployment of digital twins across industrial networks faces a critical bottleneck -> data isolation
- Strict data privacy prevents data sharing

Concept



Data:

- Synthetic IIoT dataset: 24,830 samples, 5 fault classes, 92% class imbalance
- Stratified sliding windows (SEQ_LEN = 30 timesteps, 7 sensor features)
- Distributed across 3 simulated factory clients (Non-IID split)

Federated Training:

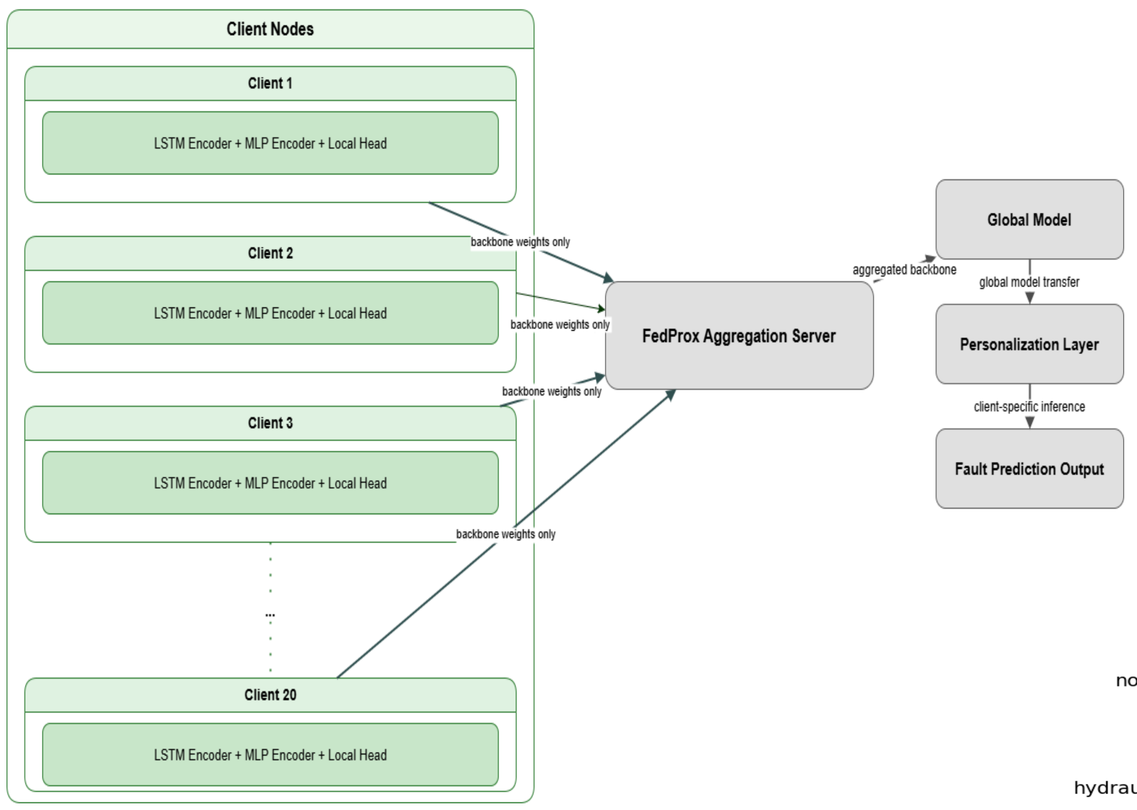
- FedProx aggregation (μ = 0.01) → stabilizes Non-IID training
- 30 communication rounds, global LSTM backbone shared
- Local heads remain private, only backbone weights aggregated

Personalization:

- Frozen global backbone transferred to each client
- Fine-tuned local classification head (LR = 1e-4, 5 epochs)
- Prevents catastrophic forgetting of rare fault classes

Architecture

F-MTD System Architecture



- Local Training at the Edge (Client Nodes)
- Secure Communication (Backbone Weights Only)
- Global Aggregation (FedProx Server)
- Personalization & Fault Prediction (Return to Client)

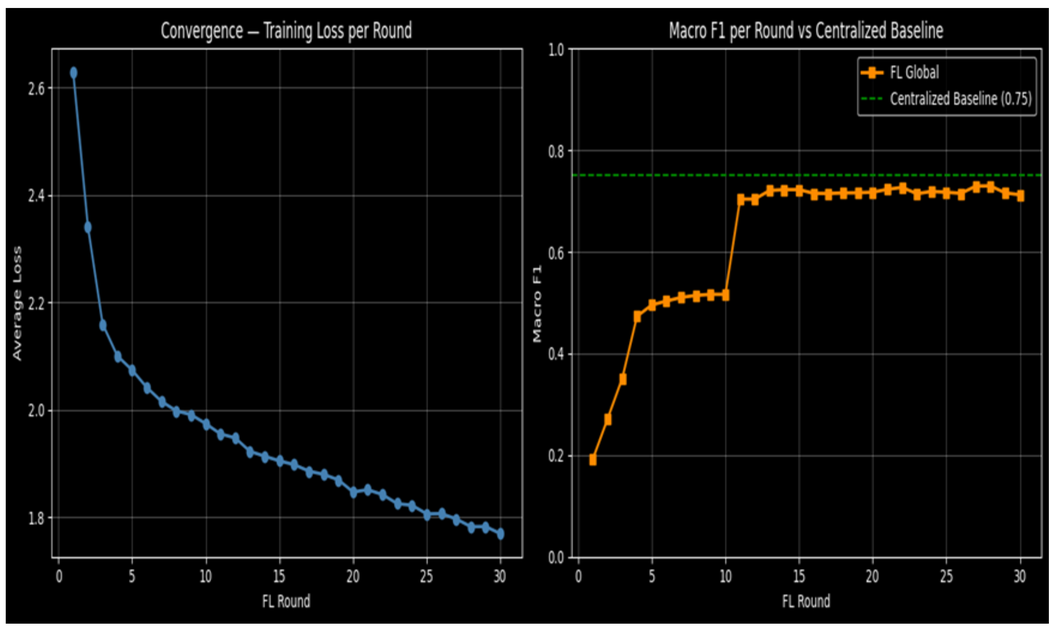
Task

- Development of a **Federated Multimodal Digital Twin (F-MDT)** framework for privacy-preserving fault detection across distributed IIoT nodes
- Implementation and evaluation of a **FedProx-based LSTM pipeline** with personalized local heads on a class-imbalanced fault dataset

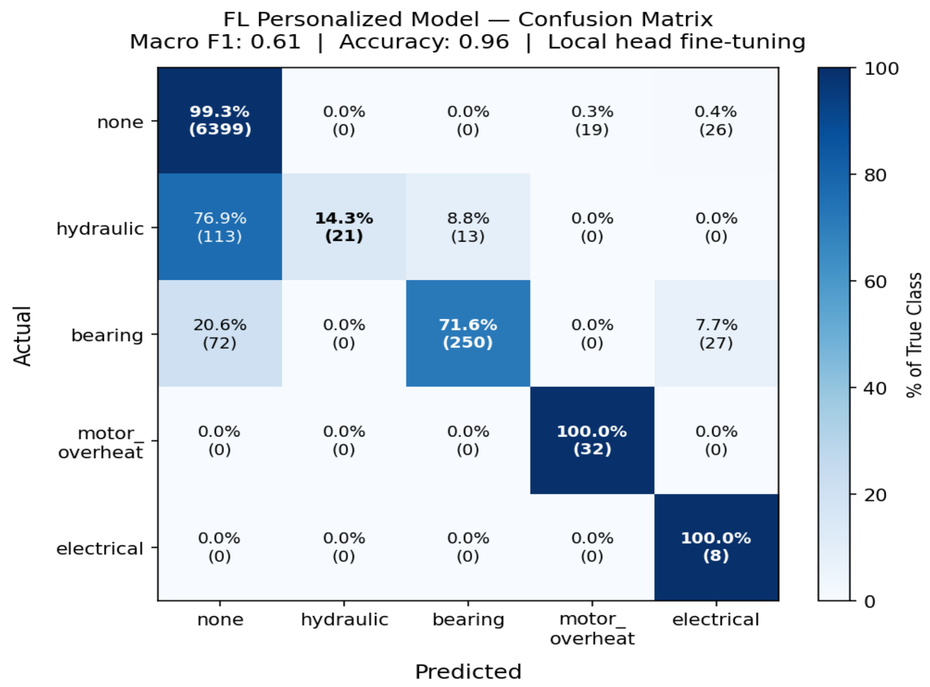
Result

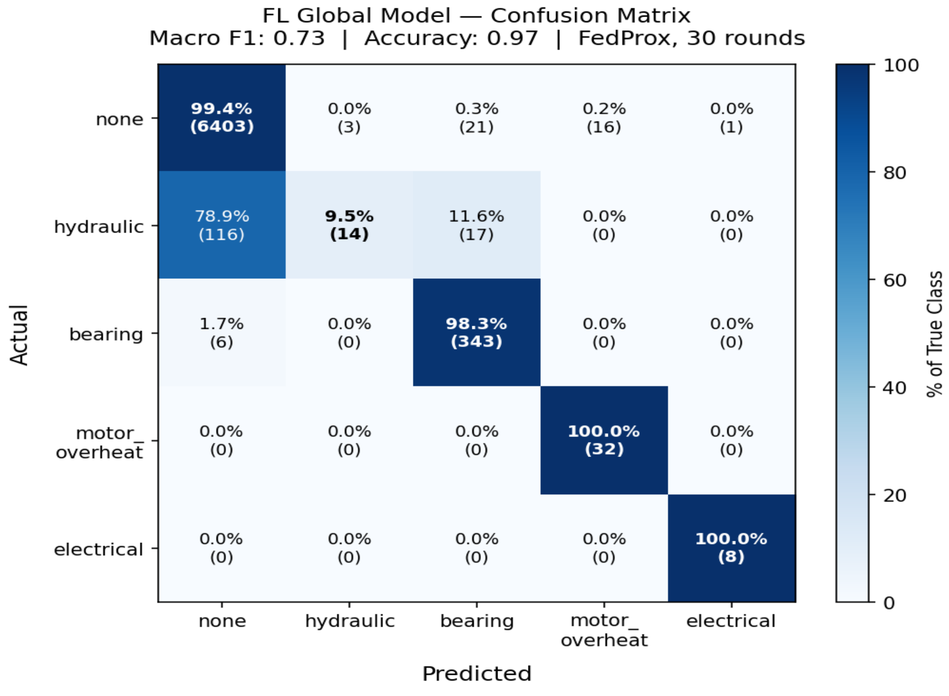
Simulation Result:

- Convergence - Training Loss per Round
- Macro F1 per Round vs Centralized Baseline



Confusion Matrices - Fault Detection Accuracy





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